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(54) **ELECTRIC VEHICLE WHEEL MOUNTED GENERATOR**

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B60L 7/14 (2006.01)

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(58) **Field of Classification Search**
CPC B60K 7/0007; H02K 7/1846
See application file for complete search history.

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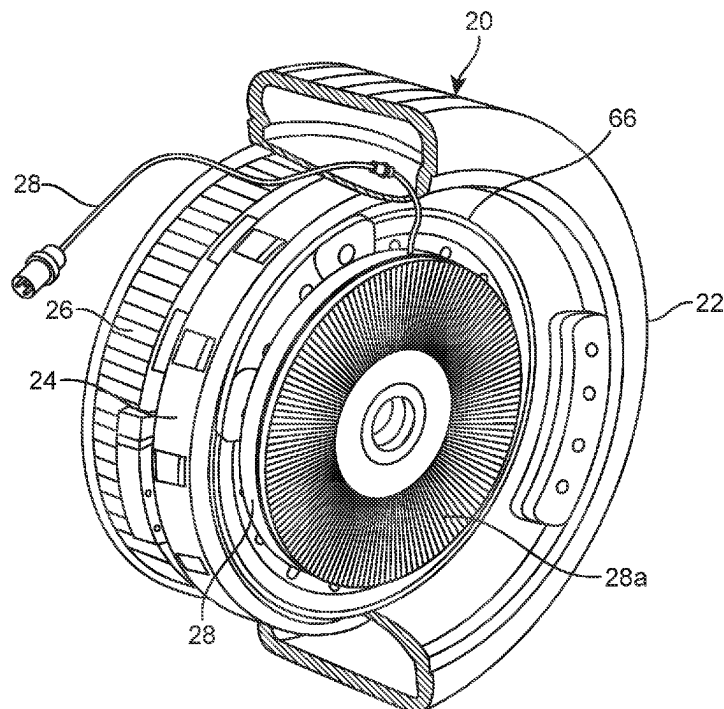
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(57) **ABSTRACT**

An electric vehicle wheel mounted generator including a vehicle wheel assembly connected to a battery assembly in the inner side bottom of an electric vehicle assembly. The vehicle wheel assembly includes vehicle wheel members with a generator attached therein which include stator coils and magnetic coils to provide electricity while electric vehicle assembly rolls. The battery assembly includes a battery which is provided with electricity by the generator with wires. Battery is connected to a switch that allows to change the power supply for the electric vehicle assembly as an external battery. Electric vehicle assembly includes an electric vehicle which has vehicle brakes attached to the vehicle wheel members. Generator saves the current in the battery and is activated by the switch when required a vehicle battery does not have enough battery.

5 Claims, 4 Drawing Sheets



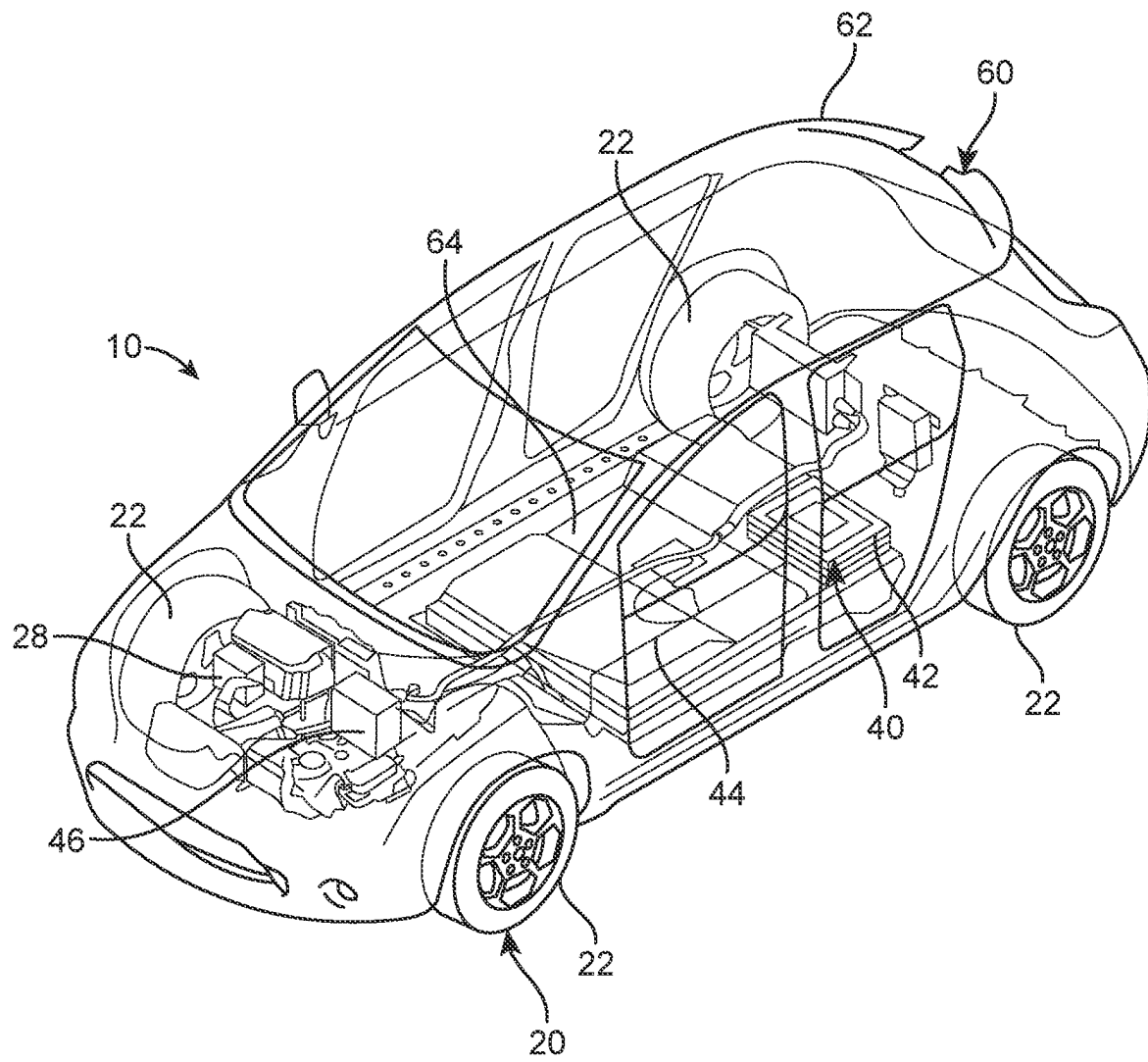


FIG. 1

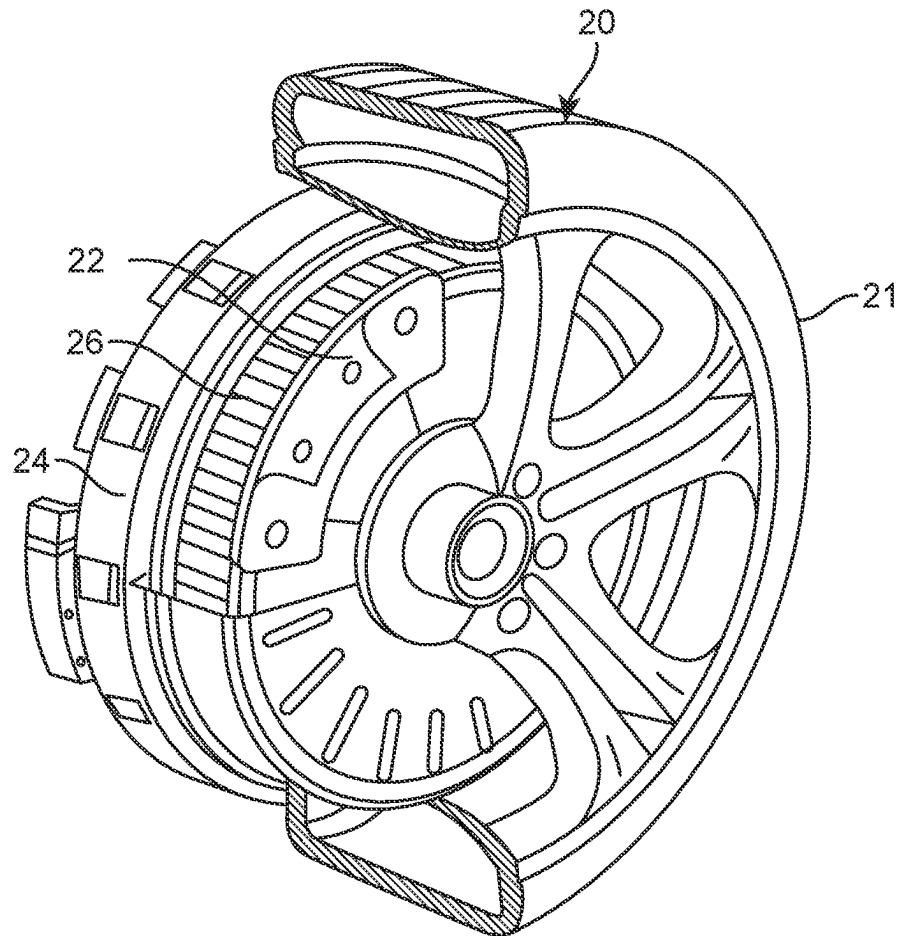


FIG. 2

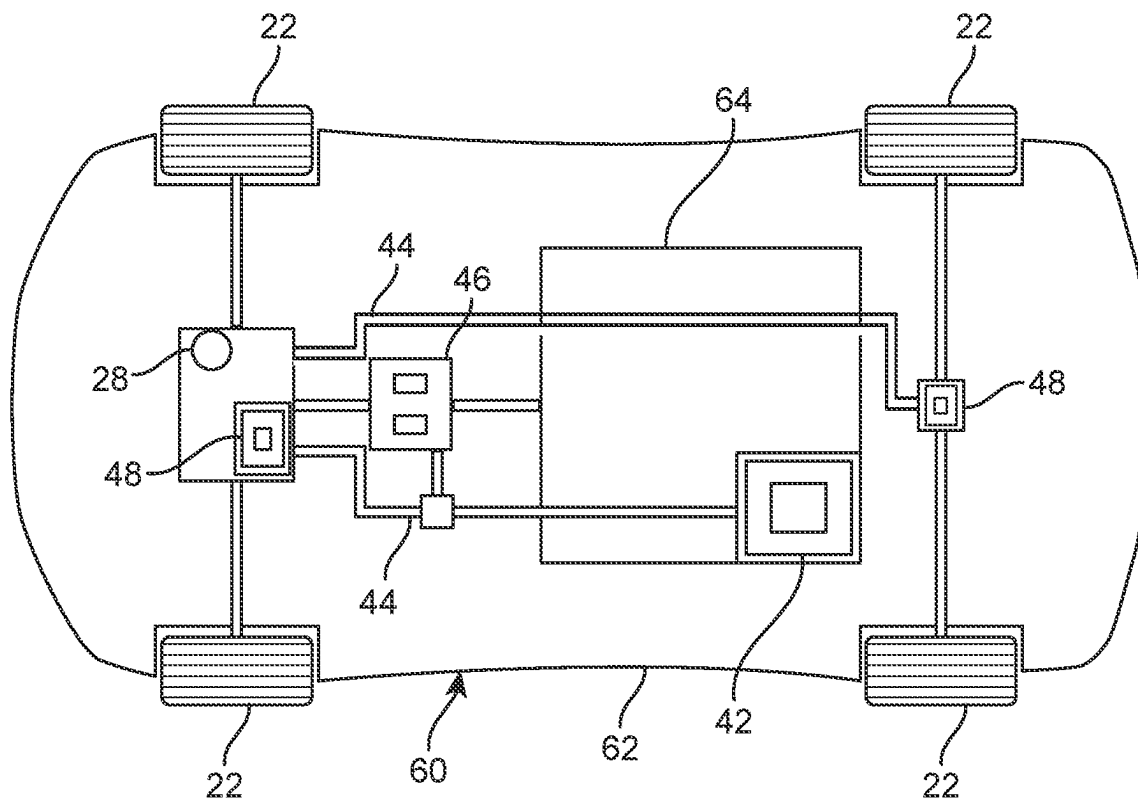


FIG. 3

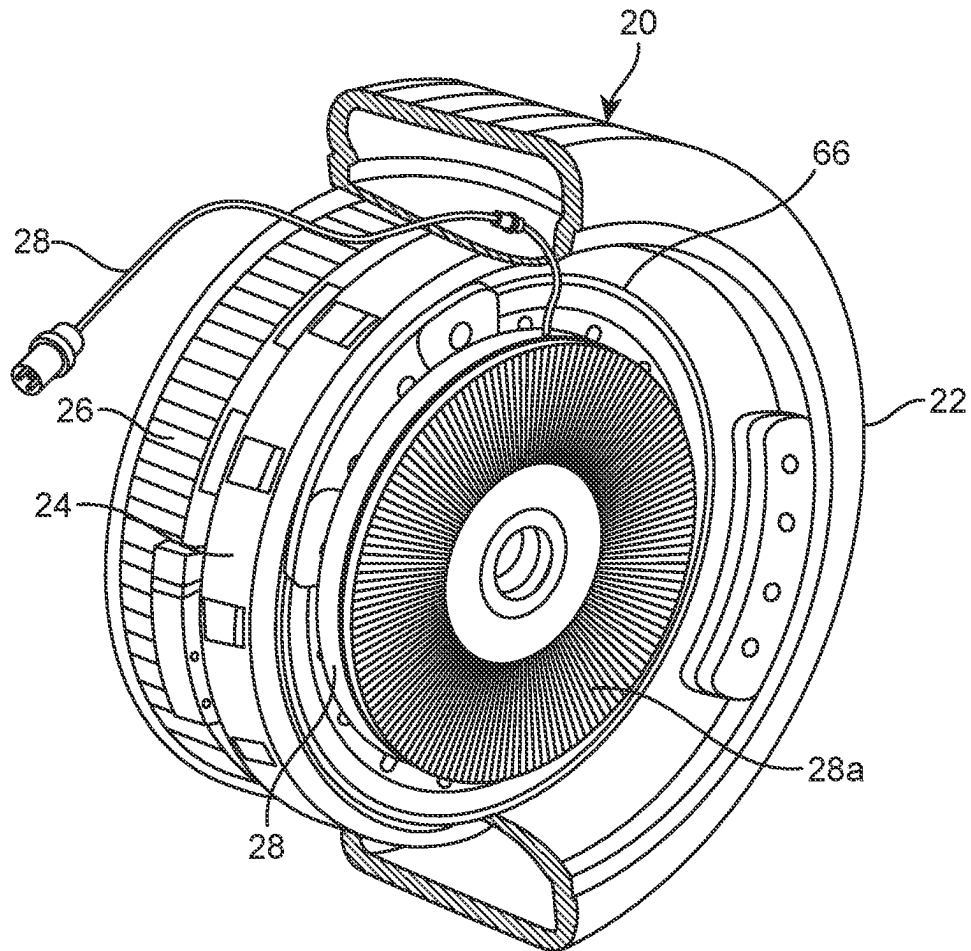


FIG. 4

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ELECTRIC VEHICLE WHEEL MOUNTED GENERATOR

1. Field of the Invention

The present invention relates to an electric vehicle wheel mounted generator and, more particularly, to an electric vehicle wheel mounted generator that includes magnetic coils placed at the perimeter of the inside of the vehicle wheel in addition with a stator attached to the vehicle brakes that creates electricity during rotation of wheels to accumulate that electricity to a supplemental battery to give a longer duration of battery to vehicle between charges.

2. Description of the Related Art

Several designs for an electric vehicle wheel mounted generator have been designed in the past. None of them, however, include magnetic coils attached to the perimeter of the inside of wheels and a stator mounted to vehicle brakes that produce electricity stored in a supplemental battery that can be configured to be used as external battery from main battery system of vehicle.

Applicant believes that a related reference corresponds to U.S. Pat. No. 4,761,577 issued for a wheel mounted electricity generator for a vehicle. Applicant believes that another related reference corresponds to U.S. Pat. No. 8,853,869 issued for a self-charging electric vehicle comprising wheel mounted magnetic coil generators which supply electricity to the batteries when depleted. None of these references, however, teach of an electricity generator system for and electric vehicle wherein the generators are located at each wheel comprising a magnetic coil located around the perimeter of the inside of the vehicle wheel and a stator mounted to the vehicle brakes wherein the rotation of the wheel creates electricity which can be stores in a supplemental battery for extending the range of the vehicle in between charges.

Other documents describing the closest subject matter provide for a number of more or less complicated features that fail to solve the problem in an efficient and economical way. None of these patents suggest the novel features of the present invention.

III. SUMMARY OF THE INVENTION

It is one of the objects of the present invention to provide magnetic coils located around the perimeter of the inside of vehicle wheels that generate electricity during rotation.

It is another object of this invention to provide a cleaning system by bristles attached to said system to dislodge and clean debris from magnets to deliver balance in wheels due clean magnets.

It is still another object of the present invention to provide an extended additional battery system for reaching longer distances in an electric vehicle.

It is yet another object of this invention to provide such a device that is inexpensive to implement and maintain while retaining its effectiveness.

Further objects of the invention will be brought out in the following part of the specification, wherein detailed description is for the purpose of fully disclosing the invention without placing limitations thereon.

IV. BRIEF DESCRIPTION OF THE DRAWINGS

With the above and other related objects in view, the invention consists in the details of construction and combi-

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nation of parts as will be more fully understood from the following description, when read in conjunction with the accompanying drawings in which:

FIG. 1 represents an operational view of the present invention 10 wherein is shown the vehicle wheel assembly 20 connected to the battery assembly 40 by wires 44.

FIG. 2 shows a perspective view of the vehicle assembly 20 wherein one of the vehicle wheel members 22 includes one of the stator coils 24 placed on one of the vehicle wheel members 22 including one of the magnetic coils 26.

FIG. 3 illustrates a diagram of the vehicle wheel assembly 20 with the battery assembly 40 attached to the inner bottom portion of the vehicle assembly 60.

FIG. 4 is a representation of a perspective sectional view of the vehicle wheel assembly 20 wherein a pickup coil 26 is connected to one of the vehicle brakes 66. Cleaners 28a are attached to the generator 22.

V. DETAILED DESCRIPTION OF THE EMBODIMENTS OF THE INVENTION

Referring now to the drawings, where the present invention is generally referred to with numeral 10, it can be observed that it basically includes a vehicle wheel assembly 20, a battery assembly 40 and an electric vehicle assembly 60. It should be understood there are modifications and variations of the invention that are too numerous to be listed but that all fit within the scope of the invention. Also, singular words should be read as plural and vice versa and masculine as feminine and vice versa, where appropriate, and alternative embodiments do not necessarily imply that the two are mutually exclusive.

The vehicle wheel assembly 20 includes vehicle wheel members 21, generators 22, wheel spokes 22a, stator coils 24, magnetic coils 28, cleaner 28a, bristles 28b. In a suitable embodiment each of the vehicle wheel members 21 may be a housing for each of the generators 22. Vehicle wheel members 21 may be made of a rubber material with steel. In one embodiment vehicle wheel members 21 may have attached stator coils 24 to each of the vehicle wheel members 21 including also magnetic coils 26 considering further the electric vehicle assembly 60. Vehicle wheel members 21 may be configured to rotate and create current to provide to the battery assembly 40 by the generators 22. Vehicle wheel members 21 may be placed in a normal wheel car configuration. Generators 22 may be attached to the inner of the spokes of each of the vehicle wheel members 21. In a suitable embodiment, generators 22, stator coils 24 and magnetic coils 26 may be made of a steel material. Nevertheless, in other embodiments generators 22 may be made of a metal material, stainless steel material, copper material, cobalt material, nickel material, aluminum material or any other material that may present magnetic properties may be suitable for generators 22, stator coils 24 and magnetic coils 26. Generators 22 may be configured to provide current to the battery assembly 40 which may be produced by the stator coils 24 and the magnetic coils 26 which are placed on each of the vehicle wheel members 21. In a preferred embodiment, generators 22 may have a body that conforms with the body of the vehicle wheel members 21.

As best observed in FIG. 2, one of the generators 22 may be attached in the inner side of one of the vehicle wheel members 21. Generators 22 may be constructed by stator coils 24 and magnetic coils 26. In one embodiment generator, 22 may be a type of DC series motor. In other embodiments generators 22 may be a brushless DC motors, Permanent Magnet Synchronous Motor, Three Phase AC

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induction Motors, Switched Reluctance Motors, or any other variation thereof. Stator coils **24** may be a housing for the magnetic coils **26**. It is to be considered that each of the generators **22** may have each of the stator coils **24** wherein each of the magnetic coils **26** is fastened thereof which produces electrical current when rotating. Stator coils **24** may be configured to convert the rotating magnetic field produced when magnetic coils **26** rotate to electric current. Stator coils **24** may be suitable to have a body that conforms with the shape of the body of each of the vehicle wheel members **21**. In a suitable embodiment, each of the stator coils **24** may be attached to each of the spokes of the vehicle wheel members **21**. Magnetic coils **26** may be suitable to have a shape that conforms with the shape of each of the vehicle wheel members **21**.

As best observed in FIG. 4, one of the pick up coils **28** may be attached to the magnetic coils **26**. In a suitable embodiment, pick up coils **28** may present a wire body with a cylindrical body. In a preferred embodiment, pick up coils **28** may be configured to allow the battery **42** to be charged with the current obtained by the generators **22**. In one embodiment, pick up coils **28** may be made of a metal material. Nevertheless, in other embodiments, pick up coils **28** may be made of a stainless-steel material, copper material, cobalt material, nickel material, aluminum material or any other material. Cleaners **28a** may have a body that conforms with each of the spokes of the vehicle wheel members **21**. Cleaners **28a** may allow to remove metal debris from the magnetism that is collected by the generators **22**. In a suitable embodiment, cleaners **28a** may be attached next to each of the vehicle wheel members **21**. In one embodiment cleaner **28a** may be made of a rigid rubber material. Nevertheless, other materials like polymer, plastic, copper, or any other material may be suitable for cleaners **28a**.

The battery assembly **40** includes a battery **42**, wires **44**, a switch **46**, amplifier **48** and diodes **48**. In a suitable embodiment battery **42** may have a square prism body as best observed in FIG. 1. Battery **42** may be attached to an inner portion of the electric vehicle assembly **80**. In one embodiment battery **42** may be placed in a bottom portion of the electric vehicle assembly **80**. Battery **42** may be configured to be activated when electric vehicle assembly **80** leaks from energy. It is to be considered that battery **42** may be provided with current by the generators **22** of each of the vehicle wheel members **21**. It is to be considered that battery **22** may be a type of lithium-ion battery considering materials, lithium, nickel, and cobalt. Nevertheless, in other embodiments, materials like lithium-iron-phosphate may be suitable for battery **42**. Battery **42** may include wires **44** which are connected to the switch **46** amplifier **48** and diodes **48**. In a preferred embodiment, wires **44** may allow the generator **22** to provide electricity to the battery **42**. Wires **44** may have a large thin cylindrical body. Wires **44** may be made of aluminum. Nevertheless, in another embodiment, wires **44** may be made of a copper material. As best shown in FIG. 3, wires **44** may be connected in the inner bottom portion of the electric vehicle assembly **60**. In a suitable embodiment switch **46** may be placed in the inner bottom of the electric vehicle assembly **60** connected by the wires **44b**.

The switch **46** may have a rectangular prism body. It is to be understood that switch **46** may allow to change the electricity supply for the electric vehicle assembly **60**. Switch **46** may be made of a steel material with electrical

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components. Switch **46** is connected by the wires **44** which connect battery **42** and vehicle battery **64**. In a preferred embodiment, it is to be considered that switch **46** may be interlocked by the wires **44** which send the electricity from the generators **22** to the switch **46**. It is to be considered that diodes **48** may allow the current flows in one direction to avoid failures in the current system. Diodes **48** may present a square body and may be connected to each of the generators **22** by the wires **44**. Diodes **48** may be made of a silicon material. Nevertheless, in other embodiments, diodes **48** may be made of a germanium material or selenium material with additional metals added. Each of the diodes **48** may be attached to a central portion wherein the generators **22** provided the current by each of the pick up coils **28** which flows by the wires **44**,

The electric vehicle assembly **60**, includes an electric vehicle **62**, a vehicle battery **64** and vehicle breaks **66**. Electric vehicle **62** may include a vehicle battery **64** which may be integrated in an internal portion thereof. Vehicle battery **64** may be connected by the wires **44** to the switch **46** and may be deactivated to allow battery **42** to provide the electricity to electric vehicle **62** when vehicle battery **64** has a leak of electricity. In a suitable embodiment, each of the vehicle brakes **66** may be the housing for each of the generators **22** wherein stator coils **24** and magnetic coils **26** are attached thereof.

The foregoing description conveys the best understanding of the objectives and advantages of the present invention. Different embodiments may be made of the inventive concept of this invention. It is to be understood that all matter disclosed herein is to be interpreted merely as illustrative, and not in a limiting sense.

What is claimed is:

1. An electric vehicle wheel mounted generator, comprising:
 - a vehicle wheel assembly including vehicle wheel members wherein each of the vehicle wheel members have a generator comprising a stator coil from stator coils and a magnetic coil from the magnetic coils, wherein each of the vehicle wheel members includes cleaners which allow to remove debris from magnets;
 - a battery assembly including a battery wherein said battery is provided of electricity by said generators from the vehicle wheel members via wires and diodes that direct current flow; and
 - an electric vehicle assembly including an electric vehicle wherein said electric vehicle includes a battery placed in the inner bottom portion thereof as a base for said battery.
2. The electric vehicle wheel mounted generator of claim 1, wherein said generator coils and stator coils have a circular shape that conforms with the shape of a spoke of the vehicle wheel members.
3. The electric vehicle wheel mounted generator of claim 1, wherein said stator coils allows pick up coils to provide the current to said diodes.
4. The electric vehicle wheel mounted generator of claim 3, wherein said diodes receive the current from wires that are connected to the vehicle wheel members by the pick up coils.
5. The electric vehicle wheel mounted generator of claim 1, wherein said electric vehicle has vehicle brakes for holding said generators.

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